

Supplemental Material for *Degradation-Aware Unfolding Network with Sparsity-Gated Experts*

Zhen Guo, Hongping Gan, *Member, IEEE*

I. PARAMETERS TABLE

We provide a summary of the notations and parameters used throughout this paper. As detailed in Table I, the variables are categorized into basic formulations, model configurations, and module-specific parameters.

TABLE I
SUMMARY OF NOTATIONS AND PARAMETERS

Category	Symbol	Description
Basic Notations	x	Original signal/image vector.
	y	Observed measurement vector.
	A	Linear sampling matrix (forward operator).
	$A^\top$	Adjoint operator (back-projection).
	M, N	Signal and measurement dimensions ($M \ll N$).
Model Configuration	n	Total iterative stages (default $n = 5$).
	k	Index of the current iterative stage.
	C_0	Initial channel count (Task-specific).
	C_1, C_2, C_3	Channel dimensions of embedding and iterative stages.
	H, W	Image spatial resolution.
DA-SA Module	$d_1^{(k)}$	Image-level degradation component.
	$d_2^{(k)}$	Measurement-level fidelity component.
	$d^{(k)}$	Degradation-aware attention mask.
MSGate Module	$w^{(k)}$	Gating scores for expert selection.
	i^*	Index of the selected Top-1 expert.
DUSE Experts	$\lambda^{(k)}$	Stage-specific learnable penalty term.
	$\tau^{(k)}$	Stage-specific learnable regularization term.
	$\mu^{(k)}$	Intermediate estimate (gradient descent output).
	$\mathcal{D}, \bar{\mathcal{D}}$	Learnable sparse transform and its inverse.
	$\mathcal{R}(\cdot)$	Regularization/prior term (e.g., ℓ_1 -norm).
Training & Loss	$\mathcal{L}_{\text{dev}}$	Reconstruction fidelity loss.
	$\mathcal{L}_{C_v}$	Expert load balancing loss.
	η	Weighting factor for $\mathcal{L}_{C_v}$.
Cross-Stage States	$x_1^{(k)}, x_2^{(k)}$	Multi-scale feature states passed between MSGate stages (init. 0).
	$\alpha^{(k)}$	Learnable SpaRSA step-size in DUSE (init. 0.5; distinct from $x_1^{(k)}, x_2^{(k)}$).

Task-specific $\mathcal{L}_{\text{dev}}$. For ICS and CS-MRI, $\mathcal{L}_{\text{dev}} = \|\mathbf{x}^{(n)} - \mathbf{x}_{\text{gt}}\|_2^2$. For SCI, we use the Charbonnier loss $\mathcal{L}_{\text{dev}} = \sum_i \sqrt{(x_i^{(n)} - x_{\text{gt},i})^2 + \epsilon^2} - \epsilon$ with $\epsilon = 10^{-3}$ [1]. The total loss is $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{dev}} + \eta \mathcal{L}_{C_v}$ with $\eta = 10^{-3}$.

II. SAMPLING PROCESS FOR COMPRESSIVE IMAGING TASKS

Natural Image Compressive Sensing (ICS): Given an input natural image $\mathbf{x} \in \mathbb{R}^{H \times W}$ with height and width of H and W , respectively, image $\mathbf{x}$ is initially partitioned into non-overlapping blocks of size $B \times B$. In cases where the width or height of $\mathbf{x}$ is not perfectly divisible by B , zero-padding is employed to ensure uniform block sizes. These blocks are then transformed into vectors, and a sampling matrix $\mathbf{A} \in \mathbb{R}^{M \times N}$ ($M \ll N$) is applied to yield measurement vector $\mathbf{y} \in \mathbb{R}^M$. In our ICS experiments, $\mathbf{A}$ is initialized as Gaussian matrix. Let $\mathcal{F}_{\text{vec}}(\cdot) : \mathbb{R}^{W \times H} \rightarrow \mathbb{R}^{B^2}$ denote the partitioning and vectorization function, and σ represent the sampling ratio, where $M = \lfloor N \times \sigma \rfloor = \lfloor B^2 \times \sigma \rfloor$. The sampling process in ICS can be represented as follows:

$$\mathbf{y} = \mathbf{A} \mathcal{F}_{\text{vec}}(\mathbf{x}). \quad (1)$$

Compressive Sensing Magnetic Resonance Imaging (CS-MRI): CS-MRI employs a partial Fourier transform matrix as the sampling matrix $\mathbf{A} = \mathbf{U}\mathbf{F}$, where $\mathbf{U}$ represents a sub-sampling mask, and $\mathbf{F}$ corresponds to the Discrete Fourier Transform (DFT). In our CS-MRI experiments, we adopt the Pseudo Radial masks as $\mathbf{U}$. Additionally, the size of $\mathbf{U}$ matches that of the input image $\mathbf{x}$, and σ denotes the ratio between the number of measurement points M and the total number of pixels N in $\mathbf{x}$, i.e., $\sigma = \frac{M}{N}$. The sampling process of CS-MRI is mathematically represented as:

$$\mathbf{y} = \mathbf{A}\mathbf{x} = \mathbf{U}\mathbf{F}\mathbf{x}. \quad (2)$$

Snapshot Compressive Sensing (SCI): Consider a 3D hyperspectral image (HSI) $\mathbf{x} \in \mathbb{R}^{H \times W \times N_\zeta}$, where W , H , and N_ζ represent its width, height, and number of wavelengths, respectively. The process begins with the application of a pre-defined coded aperture $\mathbf{A}_\zeta \in \mathbb{R}^{H \times W}$ to modulate the captured HSI, resulting in the transformed HSI denoted as $\mathbf{x}'$:

$$\mathbf{x}'(:, :, n_\zeta) = \mathbf{x}(:, :, n_\zeta) \odot \mathbf{A}_\zeta, \quad (3)$$

where $\odot$ denotes the Hadamard product, and $n_\zeta \in [1, \dots, N_\zeta]$ denotes the spectral channels. After modulation, the modulated HSI $\mathbf{x}'$ is subjected to spatial shifts through a disperser, resulting in a transformed measurement $\mathbf{x}'' \in \mathbb{R}^{H \times (W + d(N_\zeta - 1))}$. This process induces shear and tilt effects, where d denotes the step of spatial shifting. The dispersion operation can be expressed as:

$$\mathbf{x}''(u, v, n_\zeta) = \mathbf{x}'(x, y + d(\zeta_n - \zeta_c), n_\zeta). \quad (4)$$

Here, ζ_c represents the reference wavelength, ζ_n signifies the wavelength of the n_ζ -th spectral channel, (u, v) denotes the coordinate system on the detector array, and $d(\zeta_n - \zeta_c)$ characterizes the spatial shifting offset of the n_ζ -th channel on $\mathbf{x}''$. As a result, the 2D compressed measurement $\mathbf{y} \in \mathbb{R}^{H \times (W + d(N_\zeta - 1))}$ is acquired through the following summation operation:

$$\mathbf{y} = \sum_{n_\zeta=1}^{N_\zeta} \mathbf{x}''(:, :, n_\zeta) + \mathbf{E}. \quad (5)$$

Here, $\mathbf{E}$ signifies the random image noise produced by the photon sensing detector.

III. INITIALIZATION PROCESS FOR COMPRESSIVE IMAGING TASKS

ICS: For initialization, a matrix multiplication is performed using the transpose of the sampling matrix $\mathbf{A}^T \in \mathbb{R}^{N \times M}$ and $\mathbf{y}$ to obtain a vector of image blocks. Following this, the function $\tilde{\mathcal{F}}_{\text{vec}}(\cdot) : \mathbb{R}^{B^2} \rightarrow \mathbb{R}^{W \times H}$ is applied to recover the

TABLE II
THE CONFIGURATIONS OF PRETRAINING AND FINE-TUNING ON THE ICS, CS-MRI AND SCI TASKS

Configuration	ICS		CS-MRI		SCI	
	Pretrain	Fine-tune	Pretrain	Fine-tune	Pretrain	Fine-tune
sampling matrix init	Gaussian matrix	-	Pseudo Radial mask	-	Pre-defined coded aperture	-
weight init	trunc.normal (0.2)	-	trunc.normal (0.2)	-	trunc.normal (0.2)	-
α_k init	DUSE step-size: 0.5	-	DUSE step-size: 0.5	-	DUSE step-size: 0.5	-
λ_k init	1e-3	-	1e-3	-	1e-3	-
block size	32	32	-	-	-	-
stages count	5	5	5	5	5	5
image size	96×96	96×96	256×256	256×256	256×256	256×256
basic channel count C_0	1	1	1	1	28	28
stage channel count $[C_1, C_2, C_3]$	[32, 48, 80]	[32, 48, 80]	[32, 48, 80]	[32, 48, 80]	[32, 48, 80]	[32, 48, 80]
number of heads N_h	8	8	8	8	8	8
channel per head C_h	64	64	64	64	64	64
batch size	10	10	2	2	2	2
training epochs	400	400	200	100	400	400
base learning rate	2e-4	4e-5	1e-4	4e-5	1e-4	2e-5
min learning rate	1e-6	1e-6	1e-6	1e-6	1e-6	1e-6
optimizer	AdamW	AdamW	AdamW	AdamW	AdamW	AdamW
weight decay	0.05	0.05	0.05	0.05	0.05	0.05
optimizer momentum	0.9, 0.999	0.9, 0.999	0.9, 0.999	0.9, 0.999	0.9, 0.999	0.9, 0.999
warmup epochs	5	5	5	5	5	5
warmup schedule	linear	linear	linear	linear	linear	linear
learning rate schedule	cosine annealing	cosine annealing	cosine annealing	cosine annealing	cosine annealing	cosine annealing
framework	Pytorch					
GPU	RTX 4090 24 GB					

image blocks and assemble them into the initial estimate $\mathbf{x}^{(0)}$. This initialization process can be represented as:

$$\mathbf{x}^{(0)} = \tilde{\mathcal{F}}_{\text{vec}}(\mathbf{A}^T \mathbf{y}). \quad (6)$$

CS-MRI: As for initialization, we apply the inverse DFT, denoted as $\tilde{\mathbf{F}}$, to the acquired measurement $\mathbf{y}$ to obtain the initial estimate $\mathbf{x}^{(0)}$. Therefore, the initialization of CS-MRI can be expressed as:

$$\mathbf{x}^{(0)} = \tilde{\mathbf{F}}\mathbf{y}. \quad (7)$$

SCI: Regarding the initialization, we derive $\mathbf{x}'^{(0)} \in \mathbb{R}^{H \times W \times N_\zeta}$ by repeating the measurement $\mathbf{y}$ N_ζ times along the channel dimension. Subsequently, the concatenation of $\mathbf{x}'^{(0)}$ and the 3D mask $\mathbf{A} \in \mathbb{R}^{H \times W \times N_\zeta}$ in channel dimension is inputted into a convolutional layer with a kernel size of 1×1 , yielding the initial estimate $\mathbf{x}^{(0)} \in \mathbb{R}^{H \times W \times N_\zeta}$:

$$\mathbf{x}^{(0)} = \text{Conv}_1(\text{Concat}(\mathbf{x}'^{(0)}, \mathbf{A})). \quad (8)$$

IV. DETAILED SETTINGS FOR EXPERIMENTS

Detailed training settings and model configurations for each CI task are provided in Table II.

V. ADDITIONAL EXPERIMENTS

A. Visualizations of Degradation-Aware Mask (DAM)

For CS-MRI, as shown in Fig. 1, sampling is performed in the Fourier domain using a subsampling mask, resulting in aliasing artifacts. Here, both $\mathbf{d}_1^{(k)}$ and $\mathbf{d}_2^{(k)}$ capture different aspects of this degradation: $\mathbf{d}_1^{(k)}$ emphasizes edge information, while $\mathbf{d}_2^{(k)}$ focuses on broader degraded regions. The mask also becomes more refined with stage progression, revealing increasing detail.

The SCI task, as shown in Fig. 2, involves a 3D hyperspectral image compressed into a 1D measurement via a coded aperture. This process is prone to noise-induced artifacts. Initially, both $\mathbf{d}_1^{(k)}$ and $\mathbf{d}_2^{(k)}$ capture these artifacts, but as the stages evolve, $\mathbf{d}_1^{(k)}$ emphasizes texture recovery, while $\mathbf{d}_2^{(k)}$ continues to highlight noise-affected areas. The mask becomes more precise in revealing important details as the stages progress.

B. Visualizations of Image Feature Maps

As shown in Fig. 3, we present visualizations of image features and the corresponding top-1 selection of the Deep Unfolding SpaRSA Expert (DUSE) for different images at each iteration stage across various sampling ratios. In the initial stages, DUMoE tends to capture the overall contour information of the images. However, at lower sampling ratios, block artifacts may be more prominent. Nevertheless, as the iterative stages progress, the details and texture information within the images become increasingly enriched, consequently diminishing block artifacts and resulting in high-fidelity image reconstruction. Notably, it is evident that at different sampling ratios, the refinement and enhancement of details and texture information in diverse images evolve through different experts during the iterative stages. This observation underscores the ability of DUMoE to dynamically select DUSE, facilitating iterative refinement tailored to the diverse characteristics of images during the iteration stages.

C. Parameters and FLOPs Comparison on SCI Tasks

To evaluate computational efficiency, we compare the number of parameters and FLOPs of DUMoE with several state-of-the-art SCI methods on the KAIST dataset. As shown in Tab. III, while DUMoE maintains a competitive parameter count (4.07M), it achieves a superior trade-off between reconstruction

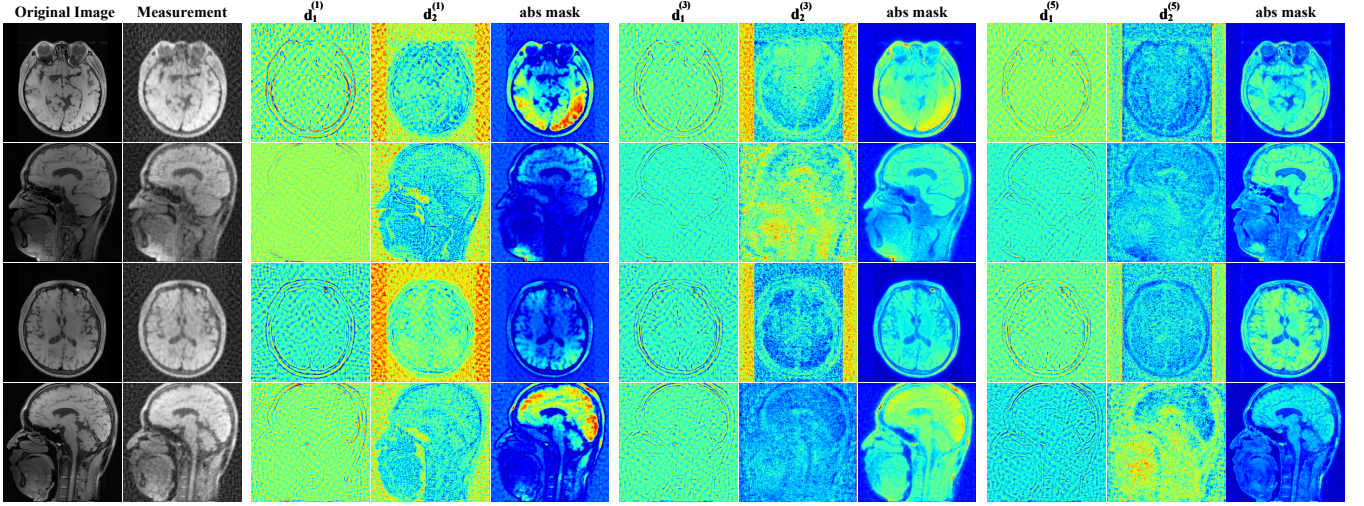


Fig. 1. The visualizations of image-level domain degradation $\mathbf{d}_1^{(k)}$, measurement-level domain degradation $\mathbf{d}_2^{(k)}$ and absolute sum of generated mask channels at stages of $k = \{1, 3, 5\}$ for CS-MRI at a sampling ratio of 0.20.

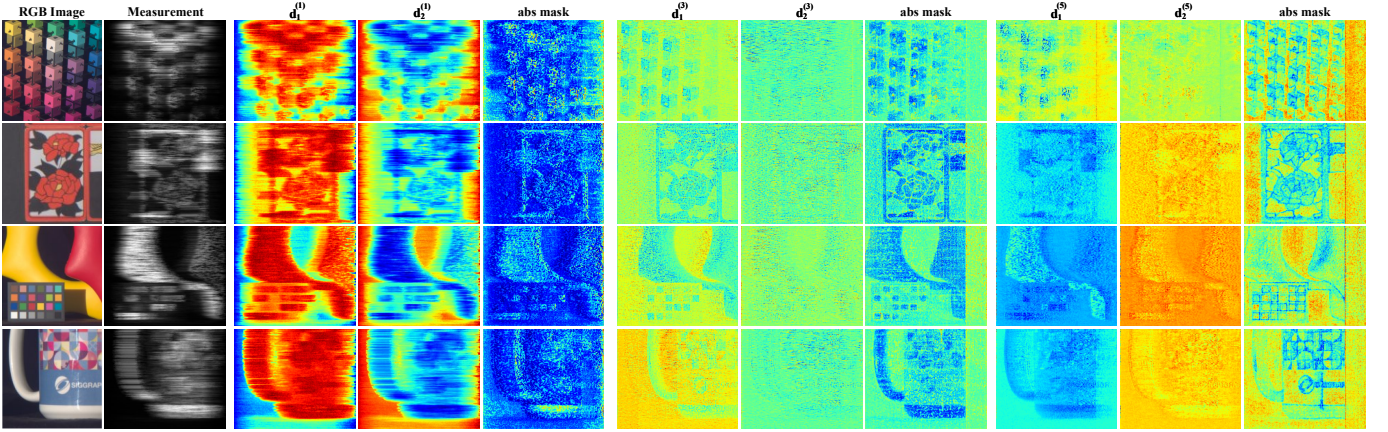


Fig. 2. The visualizations of image-level domain degradation $\mathbf{d}_1^{(k)}$, measurement-level domain degradation $\mathbf{d}_2^{(k)}$ and absolute sum of generated mask channels at stages of $k = \{1, 3, 5\}$ for SCI.

TABLE III
NUMBER OF PARAMETERS (M) AND FLOPS (G) OF DIFFERENT SCI METHODS ON THE KAIST DATASET.

Methods	ADMM-Net	Lambda-Net	TSA-Net	DGSMP	HDNet	MST-L	CST-L+	RDFNet	GAP-Net	EDUNet	DWMT	DUMoE
Params.	4.27	62.64	44.25	3.76	2.37	2.03	3.00	1.29	4.27	1.51	14.48	4.07
FLOPs	78.58	117.98	110.06	646.65	154.76	28.15	40.01	604.88	78.58	24.24	46.71	183.30

quality and computational cost compared to traditional heavy-weight models and recent efficient variants.

D. Additional CS-MRI Experiments on fastMRI Dataset

To further evaluate the performance of DUMoE, we conducted additional experiments on the fastMRI single-coil knee dataset, at a sampling ratio of 0.05 and 0.10 using Cartesian masks. As shown in Tab. IV, DUMoE achieves competitive performance in terms of average PSNR and SSIM. These results demonstrate that our proposed framework consistently maintains high-fidelity reconstruction.

TABLE IV
AVERAGE PSNR (DB) AND SSIM PERFORMANCE COMPARISONS OF DUMoE AND OTHER CS-MRI METHODS ON KNEE DATASET FROM FASTMRI USING CARTESIAN MASKS.

Methods	0.05		0.10		Avg.	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Zero-filled	23.23	0.4564	25.38	0.5314	24.31	0.4939
ISTA-Net ⁺ (CVPR 2018)	27.31	0.5594	27.82	0.6128	27.57	0.5861
RDN (AAAI 2018)	27.00	0.5630	27.32	0.5967	27.16	0.5799
DC-CNN (TMI 2018)	26.81	0.5432	27.07	0.5899	26.94	0.5666
CDDN (NeurIPS 2019)	27.45	0.5649	28.04	0.6208	27.74	0.5929
ADMM-CSNet (TPAMI 2020)	27.06	0.5512	27.43	0.5999	27.25	0.5756
HiTDUN (J-STSP 2022)	27.40	0.5630	28.02	0.6208	27.71	0.5919
PUERT (J-STSP 2022)	27.41	0.5631	27.95	0.6172	27.68	0.5902
LTWIST (TCSVT 2023)	27.23	0.5517	27.70	0.6003	27.47	0.5760
DUMoE (Our Method)	27.73	0.5754	28.89	0.6491	28.31	0.6123

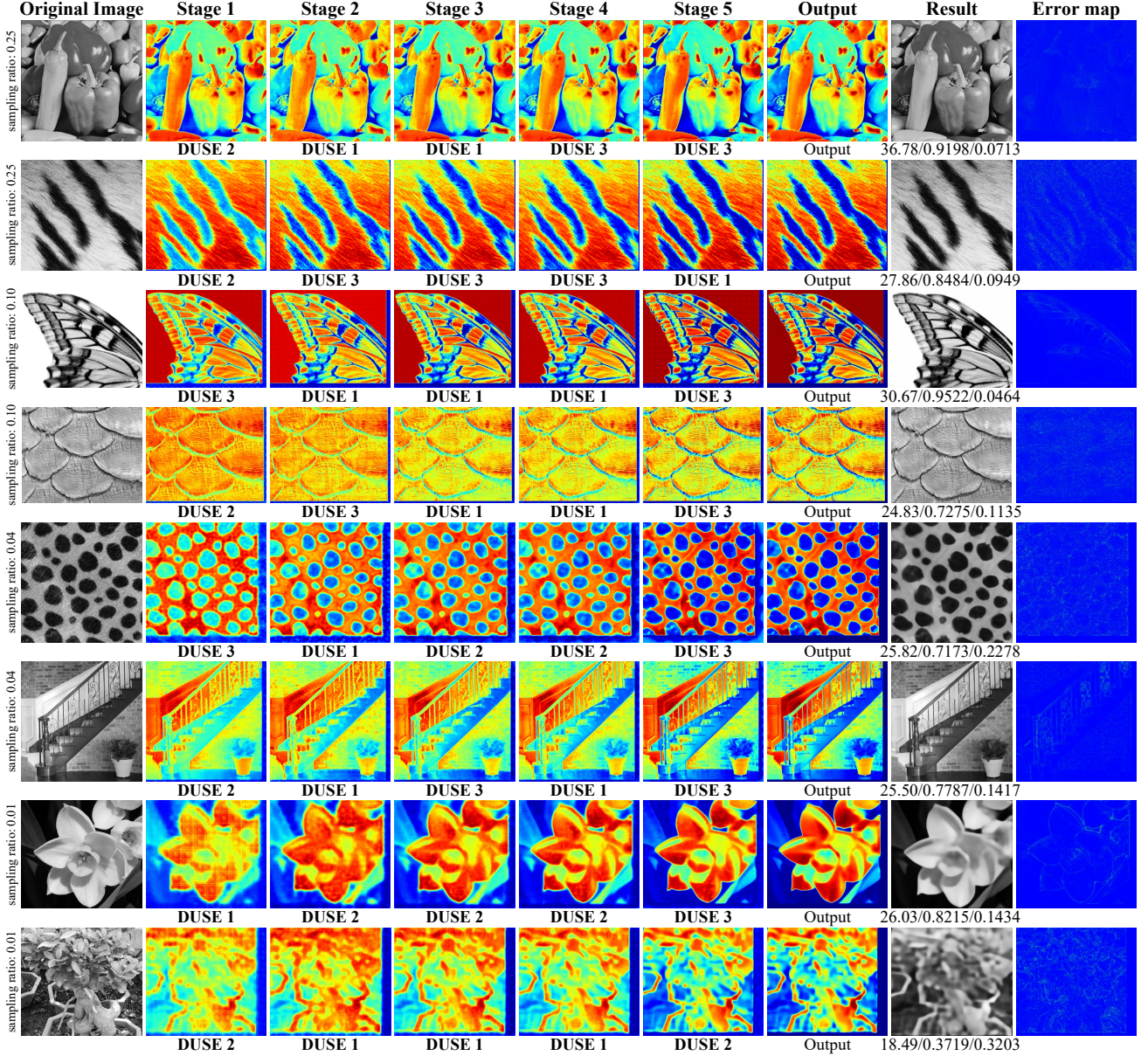


Fig. 3. Visualizations of image features and the corresponding top-1 selection of the DUSE for diverse images in each iteration stage at different sampling ratios.

E. Additional ICS Experiments on Set11 Dataset

We further provide PSNR and SSIM comparisons on the Set11 dataset under sampling ratios of 0.01, 0.04, 0.10, and 0.25. This table complements the main ICS evaluation with recent reported Set11 results. When a method does not report all sampling ratios, the corresponding entries and average values are marked as “-” to avoid unsupported interpolation or cross-protocol estimation.

VI. ALGORITHM WALKTHROUGH

Algorithm 1 in the main text summarizes the full DUMoE pipeline. The network first embeds the task-specific initialization into feature space via `Embed` and `DCAB`. At each stage k , DAM computes degradation residuals in image and measurement domains; DA-SA uses the resulting mask to

modulate self-attention values; MSGate selects one of three DUSE experts via top-1 routing; and the selected expert performs one SpARSA-style gradient-descent and proximal update. Cross-stage multi-scale states $x_1^{(k+1)}$ and $x_2^{(k+1)}$ from the U-Block are carried to the next stage (see Fig. 1 in the main text). After n stages, `PostBlock` maps features back to the image domain.

VII. OPTIMIZATION-INSPIRED DESIGN

DUMoE is motivated by SpARSA unfolding rather than a new convergence theory. Each DUSE implements one gradient-descent and proximal step, preserving the iterative interpretability of deep unfolding networks. Top-1 switch routing [31] sparsely activates one of three homogeneous experts per stage, analogous to sparse FFN activation in

TABLE V
PSNR (dB) AND SSIM PERFORMANCE COMPARISONS OF DUMoE AND OTHER ICS METHODS ON THE Set11 DATASET AT DIFFERENT SAMPLING RATIOS.

Dataset	Methods	0.01		0.04		0.10		0.25		Avg.	
		PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Set11	ReconNet (2016) [2]	17.43	0.4017	20.93	0.5897	24.38	0.7301	28.44	0.8531	22.80	0.6437
	ISTA-Net ⁺ (2018) [3]	17.45	0.4131	21.56	0.6240	26.49	0.8036	32.44	0.9237	24.49	0.6911
	SCSNet (2019) [4]	21.04	0.5562	24.29	0.7589	28.52	0.8616	33.43	0.9373	26.82	0.7785
	CSNet ⁺ (2020) [5]	20.69	0.5238	24.54	0.7445	28.12	0.8664	32.20	0.9337	26.39	0.7671
	DPA-Net (2020) [6]	18.05	0.5011	23.50	0.7205	26.99	0.8354	31.74	0.9238	25.07	0.7452
	MAC-Net (2020) [7]	18.26	0.4003	24.22	0.6982	27.68	0.8182	32.91	0.9244	25.77	0.7103
	OPINE-Net ⁺ (2020) [8]	20.02	0.5362	25.52	0.7879	29.81	0.8904	34.81	0.9514	27.54	0.7915
	AMP-Net (2021) [9]	20.20	0.5581	25.26	0.7722	29.40	0.8779	34.63	0.9481	27.37	0.7891
	BCS-Net (2021) [10]	20.81	0.5427	24.90	0.7531	29.36	0.8650	34.20	0.9408	27.32	0.7754
	COAST (2021) [11]	12.40	0.2637	23.55	0.7158	28.70	0.8609	33.96	0.9405	24.65	0.6952
	NL-CSNet (2021) [12]	21.96	0.6005	26.26	0.8108	30.05	0.8995	-	-	-	-
	CASNet (2022) [13]	21.76	0.6019	26.25	0.8118	30.29	0.9005	35.65	0.9592	28.49	0.8184
	DGUNet ⁺ (2022) [14]	22.15	0.6114	26.83	0.8230	30.93	0.9088	36.18	0.9616	29.02	0.8262
	FSOINet (2022) [15]	21.73	0.5937	26.37	0.8119	30.44	0.9018	35.80	0.9595	28.59	0.8167
	TransCS (2022) [16]	20.15	0.5066	25.41	0.7883	29.54	0.8877	35.06	0.9548	27.54	0.7844
	AutoBCS (2023) [17]	19.63	0.5605	24.73	0.7871	28.44	0.8827	33.56	0.9481	26.59	0.7946
	CSformer (2023) [18]	21.86	0.6071	26.41	0.8058	30.09	0.8925	34.99	0.9534	28.34	0.8147
	DPC-DUN (2023) [19]	18.03	0.4601	24.38	0.7498	29.42	0.8801	34.75	0.9483	26.65	0.7596
	MAPUN (2023) [20]	-	-	-	-	30.19	0.9014	35.73	0.9602	-	-
	OCTUF (2023) [21]	21.75	0.5934	26.45	0.8126	30.70	0.9030	36.10	0.9604	28.75	0.8174
	TCS-Net (2023) [22]	21.09	0.5505	25.46	0.7863	29.04	0.8834	33.94	0.9508	27.38	0.7928
	CSRNet (2023) [23]	21.40	0.5771	26.57	0.8076	29.00	0.8733	32.55	0.9293	27.38	0.7968
	LTWIST (2023) [24]	20.98	0.5469	25.62	0.7867	29.84	0.8930	35.00	0.9546	27.86	0.7953
	SODAS-Net (2023) [25]	-	-	-	-	28.89	0.8669	34.24	0.9440	-	-
	MTC-CSNet (2024) [26]	21.38	0.5664	25.45	0.7813	29.40	0.8905	34.17	0.9506	27.60	0.7972
	NesTD-Net (2024) [27]	21.40	0.5891	26.73	0.8238	30.91	0.9099	36.27	0.9622	28.83	0.8213
	PLV-CSNet (2025) [28]	21.95	0.5988	26.48	0.8126	30.49	0.9061	35.51	0.9577	28.61	0.8188
	SuperTA-Net (2025) [29]	22.29	0.5872	-	-	30.37	0.9084	35.27	0.9519	-	-
	BWCA-Net (2026) [30]	21.68	0.5949	26.35	0.8140	30.61	0.9052	35.83	0.9601	28.62	0.8186
	DUMoE (Our Method)	22.04	0.6063	27.21	0.8316	31.33	0.9132	36.35	0.9627	29.23	0.8285

Switch Transformers, while $\mathcal{L}_{C_v}$ encourages balanced expert utilization. We provide empirical evidence via ablations (main-text Table VI; main-text Table VIII) and PCA routing analysis (main-text Fig. 6), but formal convergence or stability analysis under discrete routing remains future work.

VIII. TASK ADAPTATION AND EVALUATION SCOPE

Task adaptation. DUMoE shares the same stage-wise architecture across ICS, CS-MRI, and SCI, but each task instantiates task-specific forward operators $\mathcal{F}_A$, initializers $\tilde{\mathcal{F}}_A$, basic channel count C_0 , loss function, and training schedules (Table II). We therefore describe DUMoE as a reusable deep-unfolding MoE template rather than a single model trained jointly on all tasks.

Scope of real-world evaluation. Beyond simulated benchmarks, we evaluate on fastMRI knee data (CS-MRI) and provide visual comparisons on five real CASSI scenes (SCI). Quantitative analysis under calibration error, model mismatch, and distribution shift is left as future work.

Limitations. Training and hyperparameter tuning are more involved than single-path DUNs (e.g., balancing experts via $\mathcal{L}_{C_v}$). Cross-task generalization under disparate forward models

remains open. Evaluation is still largely simulation-driven; real-scene evidence is limited to fastMRI knee metrics and visual CASSI comparisons.

REFERENCES

- [1] P. Charbonnier, L. Blanc-Feraud, G. Aubert, and M. Barlaud, “Two deterministic half-quadratic regularization algorithms for computed imaging,” in *Proc. 1st Int. Conf. Image Process.*, vol. 2. IEEE Comput. Soc. Press, 1994, pp. 168–172.
- [2] K. Kulkarni, S. Lohit, P. Turaga, R. Kerviche, and A. Ashok, “ReconNet: Non-iterative reconstruction of images from compressively sensed measurements,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 449–458.
- [3] J. Zhang and B. Ghanem, “ISTA-Net: Interpretable optimization-inspired deep network for image compressive sensing,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 1828–1837.
- [4] W. Shi, F. Jiang, S. Liu, and D. Zhao, “Scalable convolutional neural network for image compressed sensing,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 12 290–12 299.
- [5] —, “Image compressed sensing using convolutional neural network,” *IEEE Trans. Image Process.*, vol. 29, pp. 375–388, 2020.
- [6] Y. Sun, J. Chen, Q. Liu, B. Liu, and G. Guo, “Dual-path attention network for compressed sensing image reconstruction,” *IEEE Trans. Image Process.*, vol. 29, pp. 9482–9495, 2020.
- [7] J. Chen, Y. Sun, Q. Liu, and R. Huang, “Learning memory augmented cascading network for compressed sensing of images,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 513–529.

- [8] J. Zhang, C. Zhao, and W. Gao, "Optimization-inspired compact deep compressive sensing," *IEEE J. Sel. Top. Signal Process.*, vol. 14, no. 4, pp. 765–774, May 2020.
- [9] Z. Zhang, Y. Liu, J. Liu, F. Wen, and C. Zhu, "AMP-Net: Denoising-based deep unfolding for compressive image sensing," *IEEE Trans. Image Process.*, vol. 30, pp. 1487–1500, 2021.
- [10] S. Zhou, Y. He, Y. Liu, C. Li, and J. Zhang, "Multi-channel deep networks for block-based image compressive sensing," *IEEE Trans. Multimedia*, vol. 23, pp. 2627–2640, 2021.
- [11] D. You, J. Zhang, J. Xie, B. Chen, and S. Ma, "COAST: Controllable arbitrary-sampling network for compressive sensing," *IEEE Trans. Image Process.*, vol. 30, pp. 6066–6080, 2021.
- [12] W. Cui, S. Liu, F. Jiang, and D. Zhao, "Image compressed sensing using non-local neural network," *IEEE Trans. Multimedia*, vol. 25, pp. 816–830, Dec. 2021.
- [13] B. Chen and J. Zhang, "Content-aware scalable deep compressed sensing," *IEEE Trans. Image Process.*, vol. 31, pp. 5412–5426, 2022.
- [14] C. Mou, Q. Wang, and J. Zhang, "Deep generalized unfolding networks for image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 17 378–17 389.
- [15] W. Chen, C. Yang, and X. Yang, "FSOINET: Feature-space optimization-inspired network for image compressive sensing," in *Proc. IEEE Int. Conf. Acoust. Speech Signal Process. (ICASSP)*, May 2022, pp. 2460–2464.
- [16] M. Shen, H. Gan, C. Ning, Y. Hua, and T. Zhang, "TransCS: A transformer-based hybrid architecture for image compressed sensing," *IEEE Trans. Image Process.*, vol. 31, pp. 6991–7005, 2022.
- [17] H. Gan, Y. Gao, C. Liu, H. Chen, T. Zhang, and F. Liu, "AutoBCS: Block-based image compressive sensing with data-driven acquisition and noniterative reconstruction," *IEEE Trans. Cybern.*, vol. 53, no. 4, pp. 2558–2571, Apr. 2023.
- [18] D. Ye, Z. Ni, H. Wang, J. Zhang, S. Wang, and S. Kwong, "CSformer: Bridging convolution and transformer for compressive sensing," *IEEE Trans. Image Process.*, vol. 32, pp. 2827–2842, 2023.
- [19] J. Song, B. Chen, and J. Zhang, "Dynamic path-controllable deep unfolding network for compressive sensing," *IEEE Trans. Image Process.*, vol. 32, pp. 2202–2214, 2023.
- [20] —, "Deep memory-augmented proximal unrolling network for compressive sensing," *Int. J. Comput. Vis.*, vol. 131, no. 6, pp. 1477–1496, Jun. 2023.
- [21] J. Song, C. Mou, S. Wang, S. Ma, and J. Zhang, "Optimization-inspired cross-attention transformer for compressive sensing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 6174–6184.
- [22] H. Gan, M. Shen, Y. Hua, C. Ma, and T. Zhang, "From patch to pixel: A transformer-based hierarchical framework for compressive image sensing," *IEEE Trans. Comput. Imaging*, vol. 9, pp. 133–146, 2023.
- [23] Y. Zhou, Y. Chen, X. Zhang, P. Lai, L. Huang, and J. Jiang, "A lightweight recurrent learning network for sustainable compressed sensing," 2023.
- [24] H. Gan, X. Wang, L. He, and J. Liu, "Learned two-step iterative shrinkage thresholding algorithm for deep compressive sensing," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 5, pp. 3943–3956, May 2024.
- [25] J. Song and J. Zhang, "SODAS-Net: Side-information-aided deep adaptive shrinkage network for compressive sensing," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–12, 2023.
- [26] M. Shen, H. Gan, C. Ma, C. Ning, H. Li, and F. Liu, "MTC-CSNet: Marrying transformer and convolution for image compressed sensing," *IEEE Trans. Cybern.*, vol. 54, no. 9, pp. 4949–4961, 2024.
- [27] H. Gan, Z. Guo, and F. Liu, "NesTD-Net: Deep NESTA-inspired unfolding network with dual-path deblocking structure for image compressive sensing," *IEEE Trans. Image Process.*, vol. 33, pp. 1923–1937, 2024.
- [28] J. Hao, H. Bai, X. Li, M. Panic, and J. Sun, "PLV-CSNet: Projected Landweber Variant unfolding network for image compressive sensing reconstruction," *Neurocomputing*, vol. 629, p. 129723, 2025.
- [29] Z. Zhang, H. Bai, L. Stanković, J. Sun, and X. Li, "SuperTA-Net: A Supervised Transmission-Augmented Network for Image Compressive Sensing Reconstruction," in *2024 6th International Conference on Data-driven Optimization of Complex Systems (DOCS)*, 2024, pp. 586–593.
- [30] Z. Yao and J. Zhou, "BWCA-Net: Bidirectional Wavelet Cross-Attention unfolding network for image compressive sensing reconstruction," in *2026 IEEE International Conference on Image Processing (ICIP)*, 2026.
- [31] W. Fedus, B. Zoph, and N. Shazeer, "Switch Transformers: Scaling to trillion parameter models with simple and efficient sparsity," *J. Mach. Learn. Res.*, vol. 23, no. 1, pp. 120:5232–120:5270, Jan. 2022.